Gabbireddy Yasaswi Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.71
Sri Chaitanya Jnr clg	Class XII (Board of Intermediate Education AP)	2023	95.10%
KKR Gowtham E.M High School	Class X (Board of Secondary Education AP)	2021	99.50%

WORK EXPERIENCE

Data Engineering Intern | Mercedes-Benz Research & Development India

- Optimized the **recommendation system ML pipeline** by implementing **incremental training**, efficiently updating models without requiring complete retraining.
- Engineered deployment pipelines for **20 digital services** utilizing **Databricks Asset Bundles**, significantly improving deployment automation and system maintainability.
- Modernized data ingestion by migrating from Confluent Kafka to **Zerobus Ingest** on **Azure Kubernetes Service (AKS)**, successfully processing up to **10 million records** cost-effectively.
- Earned a **Letter of Appreciation** for outstanding performance, recognizing high-impact contributions to core project deliverables.

PROJECTS

Agentic RAG: Intelligent Retrieval & Inference Engine | *LangGraph, FastAPI, FAISS, MongoDB, Tavily, Ollama*

- **Engineered an adaptive, multi-agent RAG framework** using LangGraph to dynamically route user intents across vector retrieval, LLM reasoning, and Tavily live web search for highly contextual responses.
- **Implemented a self-corrective retrieval loop** with state-machine nodes to autonomously grade document relevance and rewrite ineffective queries, improving accuracy for out-of-domain prompts.
- **Deployed a scalable, end-to-end pipeline** integrating automated PDF/TXT ingestion, FAISS semantic search, MongoDB conversational memory, and FastAPI endpoints for stateful, multi-turn interactions.

Robust Zero-Shot Coreset Selection via Density-Aware Subspace Sampling (course Project)

- **Architected** a density-aware coreset selection framework using k-NN statistical filtering and subspace sampling to maintain full-dataset accuracy without labels or retraining .
- **Engineered** a robust feature preprocessing pipeline by computing distance z-scores on DINOv2 embeddings to prune outliers and prevent noise degradation .
- **Achieved** 93.44% classification accuracy on CIFAR-10 by training on a 5% data subset under high-noise conditions to outperform the standard ZCore baseline by 44% .

Online Sports Retail Revenue Optimizer | React.js, Node.js, Express.js, PostgreSQL

- Developed a **full-stack** retail platform using **React.js** and **Node.js** to dynamically optimize **product pricing** and inventory based on sales trends.
- Engineered a normalized **PostgreSQL** database utilizing **SQL stored procedures** and **triggers** to fully automate backend optimization logic.
- Implemented secure **role-based access control (RBAC)** across the frontend to seamlessly manage administrative and user privileges.

SKILLS

Languages: Python, C++, C, SQL, JavaScript, HTML, CSS, Assembly

Core CS Concepts: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems

AI & Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision,Large Language Models (LLMs), Retrieval-Augmented Generation (RAG)

Frameworks/Libraries: PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, Pandas, NumPy, Matplotlib, React.js

Databases: PostgreSQL, MySQL

Tools: Git, GitHub, Jupyter Notebook, VS Code

POSITIONS OF RESPONSIBILITY

Placement Coordinator - Office of Career Services, IIT Hyderabad

- Engaged industry recruiters to pitch student talent and expand technical placement opportunities for the graduating cohort.
- Streamlined the recruitment cycle, acting as the primary liaison between hiring companies and students to ensure a smooth interview process.
- Analyzed recruiter feedback on candidate performance to guide student preparation, directly improving overall placement success.

Internship Coordinator - Office of Career Services, IIT Hyderabad

- Pitched student profiles to industry recruiters, successfully onboarding new companies to expand the university's hiring network.
- Managed end-to-end recruitment events including pre-placement talks and interview drives ,ensuring a smooth experience for both HR teams and candidates.
- Acted as the main liaison between students and corporate partners, matching candidate skills with job requirements to improve overall placement rates.

Hospitality Coordinator - DIESTA 2025 (Interdepartmental fest)

- Managed end-to-end hospitality for 300+ participants at DIESTA
- Optimized check-in process and ensured seamless guest experience through cross-functional coordination

RELEVANT COURSES

Artificial Intelligence- Intro to AI, Artificial Intelligence, Programming for AI, Foundations of Machine Learning, Deep Learning, Reinforcement Learning, Computer Vision, Natural Language Processing

Computer Science- Introduction to Programming, Data Structures, Algorithms, DBMS, Computer Architecture, Operating Systems - 1

Mathematics - Discrete Maths, Probability and Random Variables, Matrix Theory, Applied Statistics, Convex Optimization, Differential Equations, Vector Calculus, Series of Functions, Metric Spaces

EXTRACURRICULAR

- Solved diverse DSA problems on platforms like LeetCode to enhance problem-solving skills and coding proficiency
- Participated in regular dance sessions and workshops hosted by the campus dance club.
- Volunteered with the National Service Scheme (NSS), supporting community health and welfare initiatives including the Swachh Bharat campaign and campus Walkathons.